



Spatial Interpolation — Test Results

Automated Test Documentation for Model Governance

MKM Research Labs

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1 Test Results — Spatial Interpolation (MKM-SP-001)

Tests run on **29 March 2026 at 10:29:14.**

Table 1: Test Results Summary — Spatial Interpolation (MKM-SP-001)

Total Tests	41
Passed	41
Failed	0
Skipped	0
Duration	0.01s

Table 2: Detailed Test Results — Spatial Interpolation

Test Description	Acceptance Criteria	Result	Time
Diagonal is one	—	Pass	0.000s
Off diagonal between zero and one	<code>np.all(off_diag >= 0) and np.all(off_diag <= 1)</code>	Pass	0.000s
Shape is n by n	<code>mat.shape == (n, n)</code>	Pass	0.000s
Symmetric	—	Pass	0.000s
Diagonal is one	—	Pass	0.000s
Higher base correlation increases off diagonal	<code>np.all(off_high > off_low)</code>	Pass	0.000s
Larger correlation_distance → slower exponential decay → higher off-diag.	<code>np.all(off_far >= off_near)</code>	Pass	0.000s
Off diagonal positive	<code>np.all(off > 0)</code>	Pass	0.000s
Returns square matrix	<code>mat.shape == (3, 3)</code>	Pass	0.000s
Symmetric	—	Pass	0.000s
Empty dataframe returns none	<code>build_spatial_index(df) is None</code>	Pass	0.000s
None gauge data returns none	<code>build_spatial_index(None) is None</code>	Pass	0.000s
With gauges returns kdtree	<code>isinstance(result, cKDTree)</code>	Pass	0.000s
Lines 43-45: coordinates are converted to radians.	<code>tree.data[0][1] == pytest.approx(np.pi, rel=1e-6)</code>	Pass	0.000s
Single gauge should still produce a valid KDTree.	<code>isinstance(tree, cKDTree); tree.n == 1</code>	Pass	0.000s
Lines 42-46: non-empty DataFrame builds a cKDTree.	<code>isinstance(tree, cKDTree); tree.n == 3</code>	Pass	0.000s
Equatorial degree approx 111km	<code>111.000 < d < 112.000</code>	Pass	0.000s
London paris approx 340km	<code>330.000 < d < 350.000</code>	Pass	0.000s
North south degree approx 111km	<code>110.000 < d < 112.000</code>	Pass	0.000s

Test Description	Acceptance Criteria	Result	Time
Returns float	<code>isinstance(haversine_distance(51.0, -0.1, 52.0, 0.0), float)</code>	Pass	0.000s
Same point is zero	<code>haversine_distance(51.5, -0.1, 51.5, -0.1) == pytest.approx...</code>	Pass	0.000s
Symmetry	<code>d1 == pytest.approx(d2, rel=1e-9)</code>	Pass	0.000s
All zero weights returns zero	<code>result == pytest.approx(7.5)</code>	Pass	0.000s
Closer gauge has higher weight	<code>result < 5.0</code>	Pass	0.000s
Empty list returns zero	<code>idw.interpolate_from_gauges([]) == 0.0</code>	Pass	0.000s
Equal distance gauges averages wse	<code>result == pytest.approx(3.0)</code>	Pass	0.000s
Near gauge returns wse directly	<code>result == pytest.approx(5.0)</code>	Pass	0.000s
Power one changes result	<code>r2 != pytest.approx(r1, rel=0.01)</code>	Pass	0.000s
Single far gauge	<code>result == pytest.approx(3.0)</code>	Pass	0.000s
Line 158: power parameter affects weights.	<code>r3 < r1</code>	Pass	0.000s
Line 149-150: empty input → 0.0.	<code>idw.interpolate_from_gauges([]) == 0.0</code>	Pass	0.000s
Gauge at exactly <code>min_distance=1.0</code> is NOT less than, so IDW used.	<code>result == pytest.approx(5.0)</code>	Pass	0.000s
Multiple gauges produce weighted result.	<code>2.0 < result < 6.0; result < 4.0</code>	Pass	0.000s
Line 156-157: <code>dist ; min_distance</code> → return wse.	<code>result == pytest.approx(42.0)</code>	Pass	0.000s
At gauge location returns wse directly	<code>result == pytest.approx(3.0)</code>	Pass	0.000s
Empty gauge data returns zero	<code>result == 0.0</code>	Pass	0.000s
Idw returns weighted value	<code>2.5 <= result <= 4.5</code>	Pass	0.000s
Single gauge	<code>result > 0</code>	Pass	0.000s
Lines 119-120: <code>distance ; 1m</code> → direct return.	<code>result == pytest.approx(7.0)</code>	Pass	0.000s
Line 97-98: no gauge data → 0.0.	<code>spatial.interpolate_wse(51.5, -0.1, {}, {}) == 0.0</code>	Pass	0.000s
Lines 113-124: IDW with two equidistant gauges.	<code>result == pytest.approx(2.0, rel=0.01)</code>	Pass	0.000s

1.1 Test Document History

Date	Version	Description	Author
29-Mar-2026	1.0	Initial test suite (41 tests)	D. Kelly